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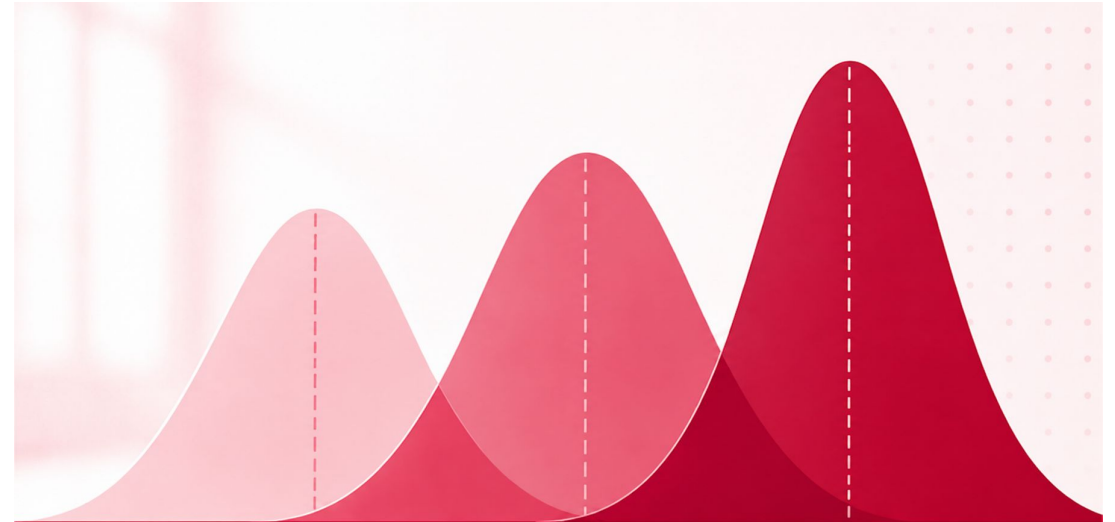
Introduction to Bayesian Inference

A Hands-on Workshop for
Early-Career Researchers

Day 2

Julius M. Pfadt & Henrik R. Godmann

17-18 September 2026



Contents

- Bayesian multi-model inference
- Mixed-effects regression in R with brms
- How to conduct and report a Bayesian analysis
- Bring your own data
- Beyond the workshop
- Conclusions

Bayesian Multi-Model Inference



...specifically

Bayesian Regression / ANOVA



Regression

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Classical Regression

- Single coefficient: $\beta_j = 0$ tested (one coefficient at a time) - just t -tests in the skies

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Classical Regression

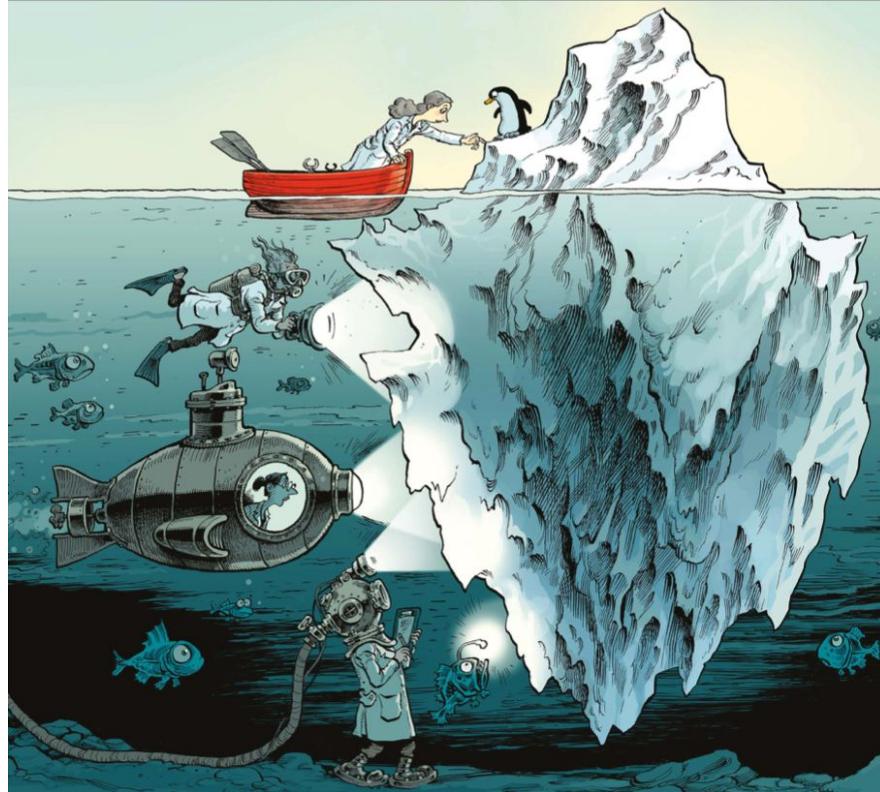
- Single coefficient: $\beta_j = 0$ tested (one coefficient at a time) - just t -tests in the skies
- Whole model (combined predictors): the F - test
- Identical interpretation: p is $P(\text{data} \mid \beta_j = 0)$, each test is conditional on the other predictors being in the model

Regression

- Test whether particular regression coefficients are zero or not
- Estimate the regression coefficients that are not zero

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2)$$

Classical Regression



Wagenmakers, E. J., Sarafoglou, A., & Aczel, B. (2022).
One statistical analysis must not rule them all. *Nature*,
605(7910), 423-425.

Bayesian Model Averaging

A Conceptual Introduction to Bayesian Model Averaging

Max Hinne

Department of Psychology, University of Amsterdam

Quentin F. Gronau

Department of Psychology, University of Amsterdam

Don van den Bergh

Department of Psychology, University of Amsterdam

Eric-Jan Wagenmakers

Department of Psychology, University of Amsterdam

THE PRIOR MODEL DEMONS ALL SHOUT ...



'INCLUDE ALL
PREDICTORS!'

'YOU NEED
ONLY THIS
PREDICTOR!'

... BUT AFTER THEY GORGE ON DATA ...



... SOME POSTERIOR MODEL DEMONS BECOME POWERFUL, AND OTHERS WITHER AWAY ...



Bayesian Model Averaging

- Demons never die
- Conclusions are based on a weighted combination of demons, with weights determined by the data

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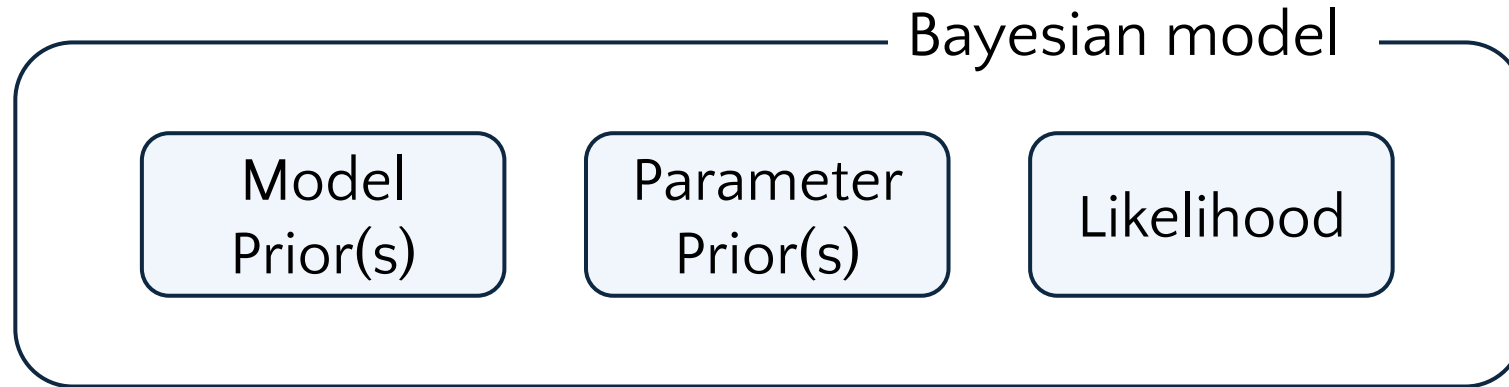


Adapted from DW

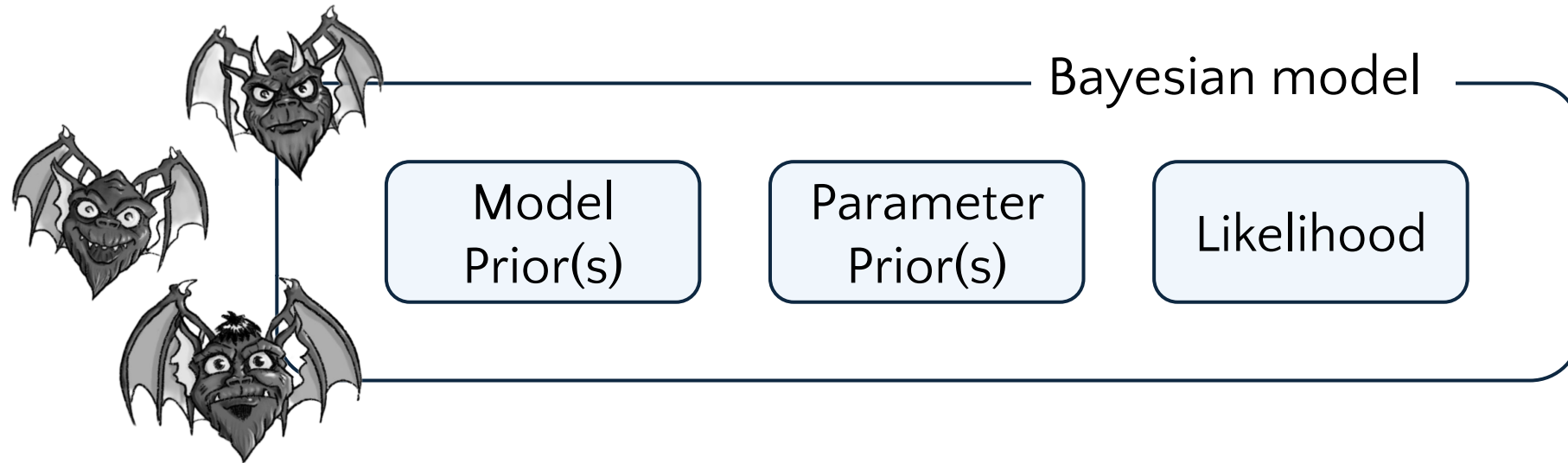
Bayesian Model Averaging

- Demons never die
- Conclusions are based on a weighted combination of demons, with weights determined by the data
- Thus:
 - **no a priori model selection**, but a selection of predictors
 - **no a posteriori model selection**, but combined inference

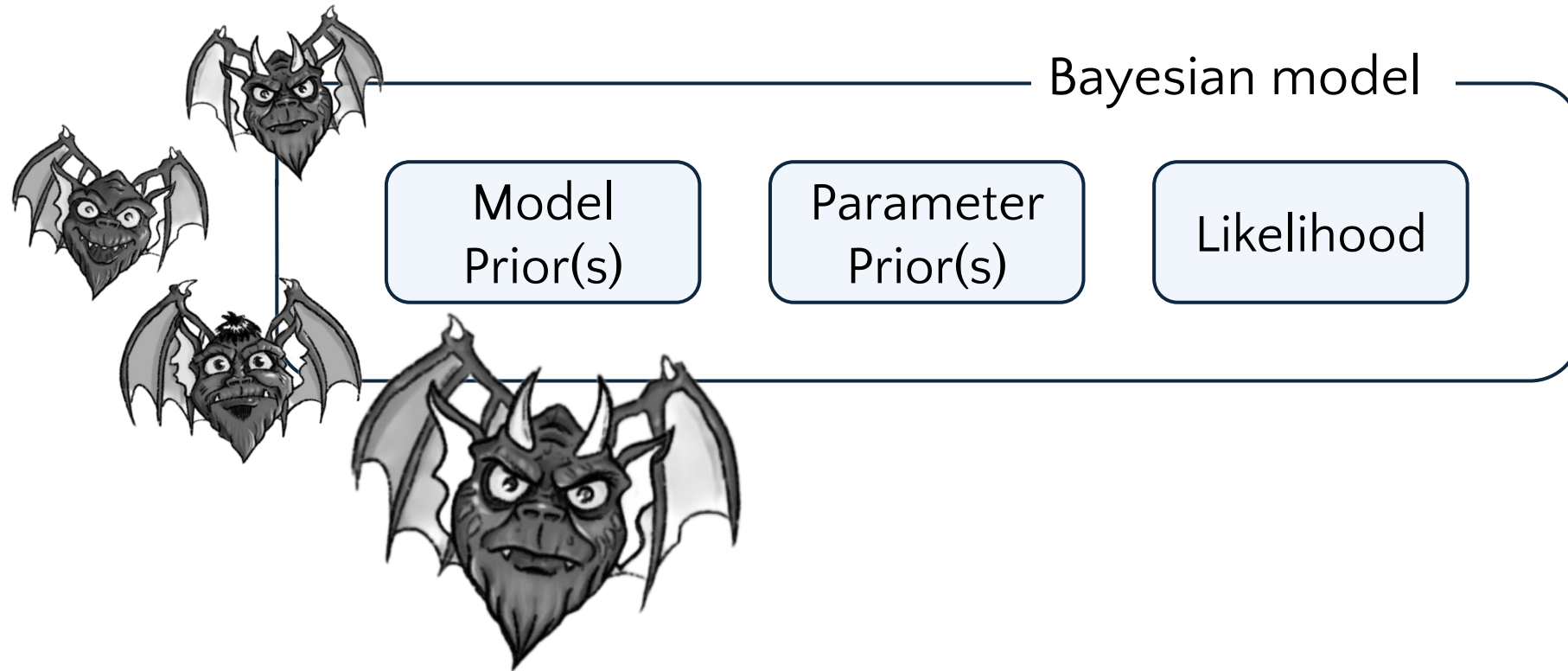
Bayesian Regression – Theory



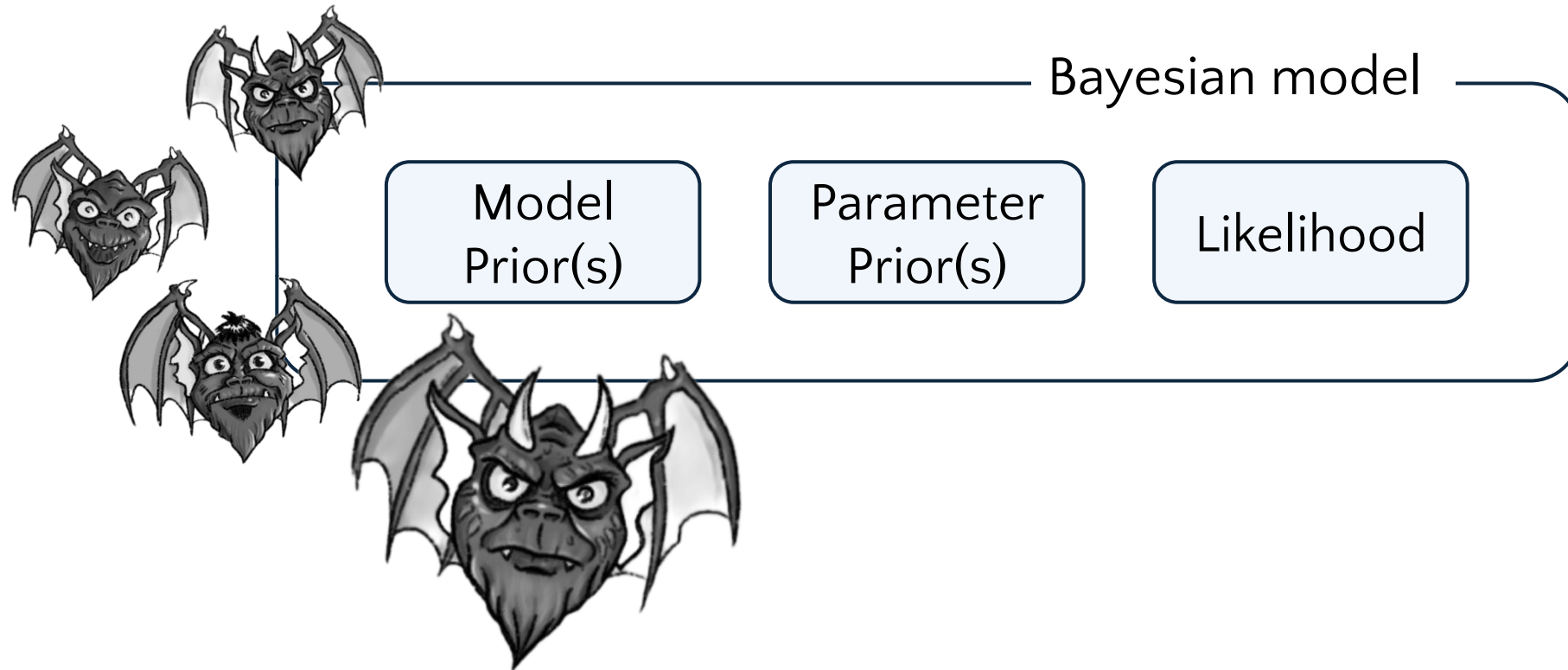
Bayesian Regression – Theory



Bayesian Regression – Theory

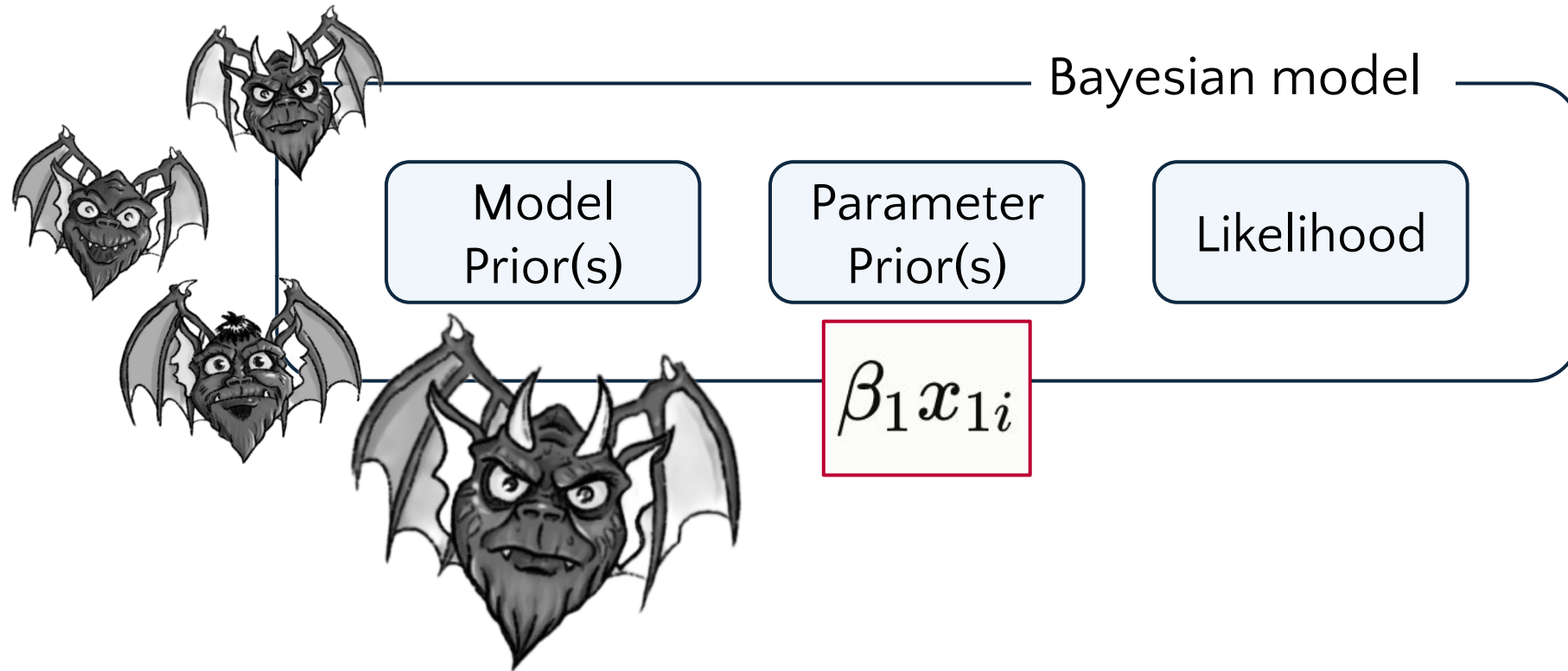


Bayesian Regression – Theory



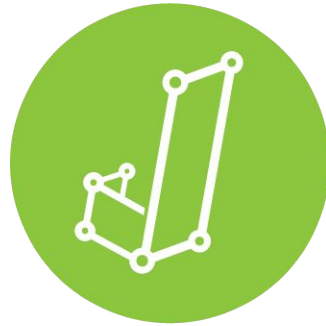
(3 predictors: $2^3 = 8$ models)

Bayesian Regression – Theory



Bayesian Regression – Theory

Model
Prior(s)



Later in JASP

Bayesian Regression – Theory

Parameter
Prior(s)

JZS - The default prior on each of the regression coefficient(s) is the same Cauchy on a standardized effect we used for the t -test, just multivariate. **nothing new.**

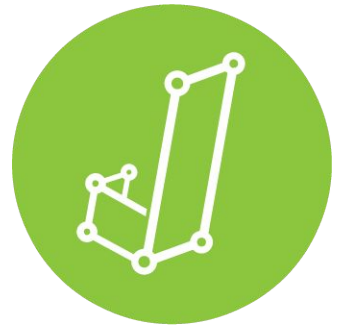
Bayesian Regression – Theory

Parameter
Prior(s)

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(... this is the standard 'medium' JZS scale, half the width we used in the *t*-test because effects are expected to be smaller when several predictors share the prior; 'wide' (0.5) and 'ultrawide' (0.707) loosen it, and should not move the results too strongly!)

Bayesian Regression – Example Part 1



Let's do a **Bayesian Regression analysis** together.

Example data set:

World Happiness (JASP data library)

Database ▶

Data Library ▶



World Happiness

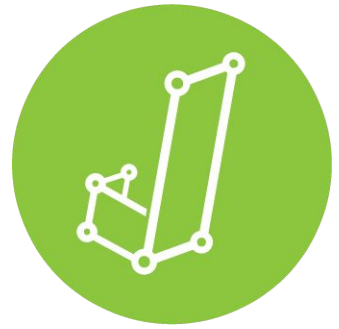
National happiness scores and various life factors of 155 countries reported in 2017.

The example JASP file demonstrates the use of Bayesian linear regression.

Data from Helliwell, Layard, & Sachs (2017).



Bayesian Regression – Exercise 2



Your turn!

Exercise data set: **Coffeeshops in the Netherlands**

use **CoffeeData.csv**. Please do a Bayesian Regression Analysis. We are interested in the association between:

Dependent

Predictors



room_temperature	distance_competitor	avg_coffee_price	floor_space	daily_revenue
27.6	0.55	4.74	144	398
17.5	2.13	4.66	104	368
18.1	0.76	2.02	117	284

Bayesian Regression – Further Reading

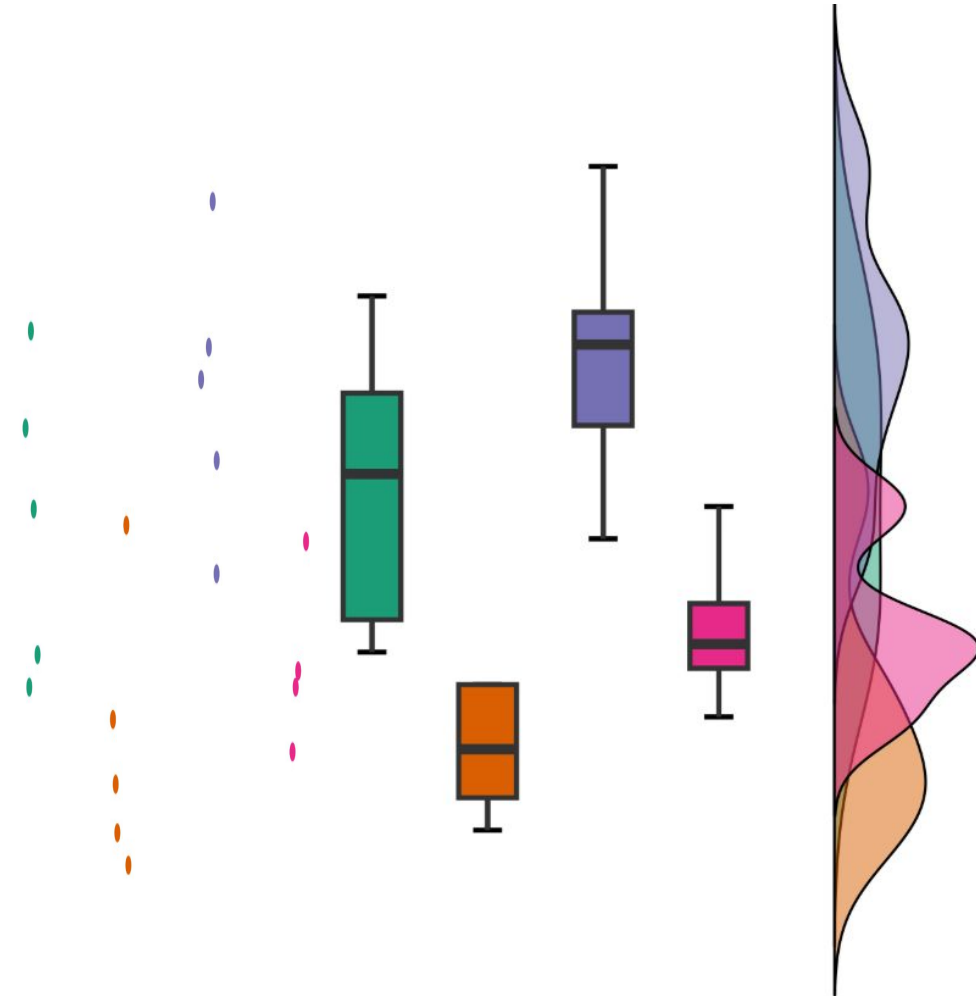
Hinne, M., Gronau, Q. F., van den Bergh, D., & Wagenmakers, E. J. (2020). A conceptual introduction to Bayesian model averaging. *Advances in Methods and Practices in Psychological Science*, 3(2), 200-215.

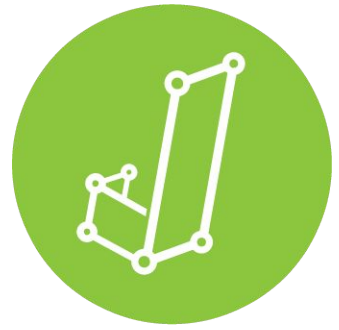
Bergh, D. V. D., Clyde, M. A., Gupta, A. R. K. N., de Jong, T., Gronau, Q. F., Marsman, M., ... & Wagenmakers, E. J. (2021). A tutorial on Bayesian multi-model linear regression with BAS and JASP. *Behavior research methods*, 53(6), 2351-2371.



ANOVA – Differences to Regression

- Categorical predictor with more than 2 levels (2 = t -test, continuous = regression)
- Uniform model prior as default
- Post-hoc tests with a correction for multiple testing





Bayesian ANOVA – Example Part 1

Let's do a **Bayesian ANOVA analysis** together.

Example data set: **Pain threshold** (JASP data library)

Bayesian ANOVA – Further Reading

van den Bergh, D., Van Doorn, J., Marsman, M., Draws, T., Van Kesteren, E. J., Derks, K., ... & Wagenmakers, E. J. (2020). A tutorial on conducting and interpreting a Bayesian ANOVA in JASP. *L'Année psychologique*, 120(1), 73-96.



Wagenmakers, E. J., Love, J., Marsman, M., Jamil, T., Ly, A., Verhagen, J., ... & Morey, R. D. (2018). Bayesian inference for psychology. Part II: Example applications with JASP. *Psychonomic bulletin & review*, 25(1), 58-76.